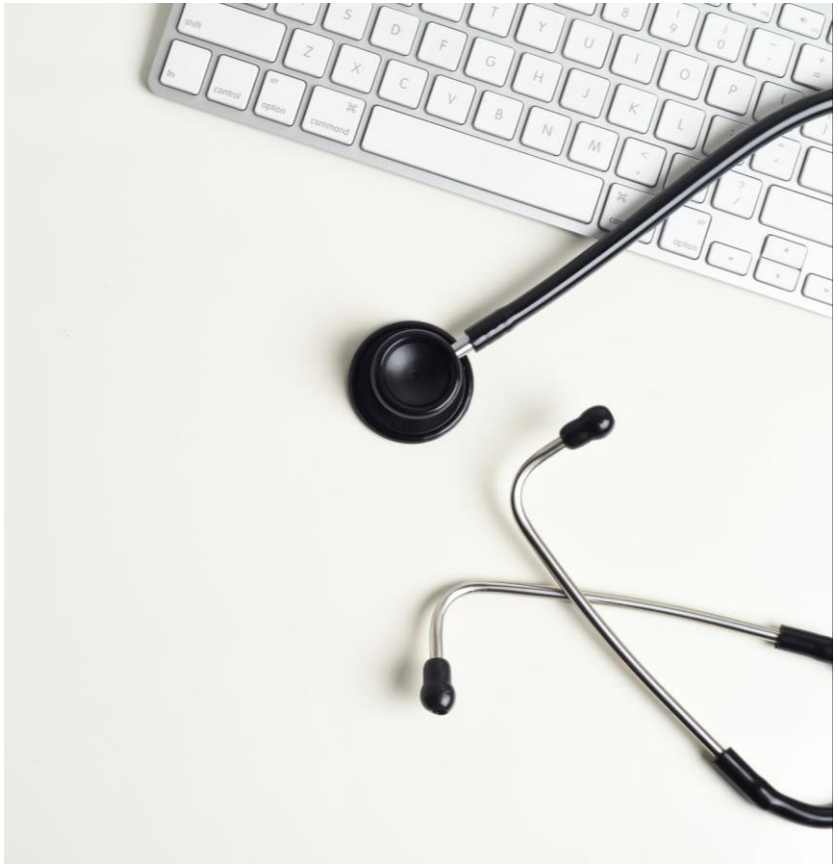
Neuromuscular conditions: : Muscle disorders

Prof Isaac Sorinola

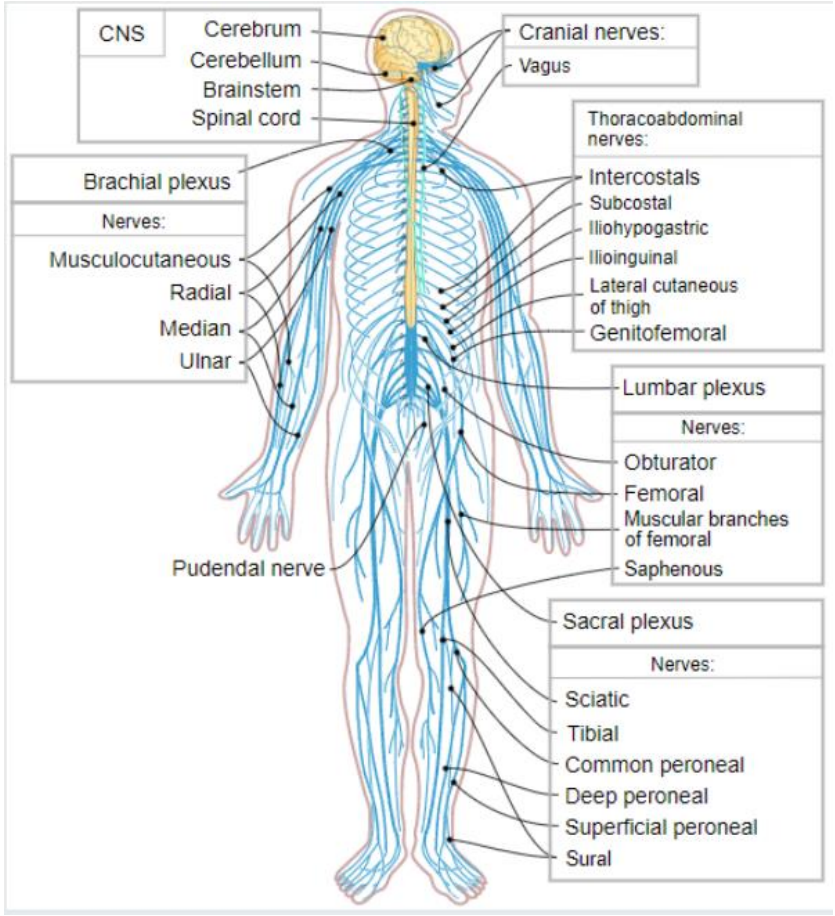
Learning Outcomes

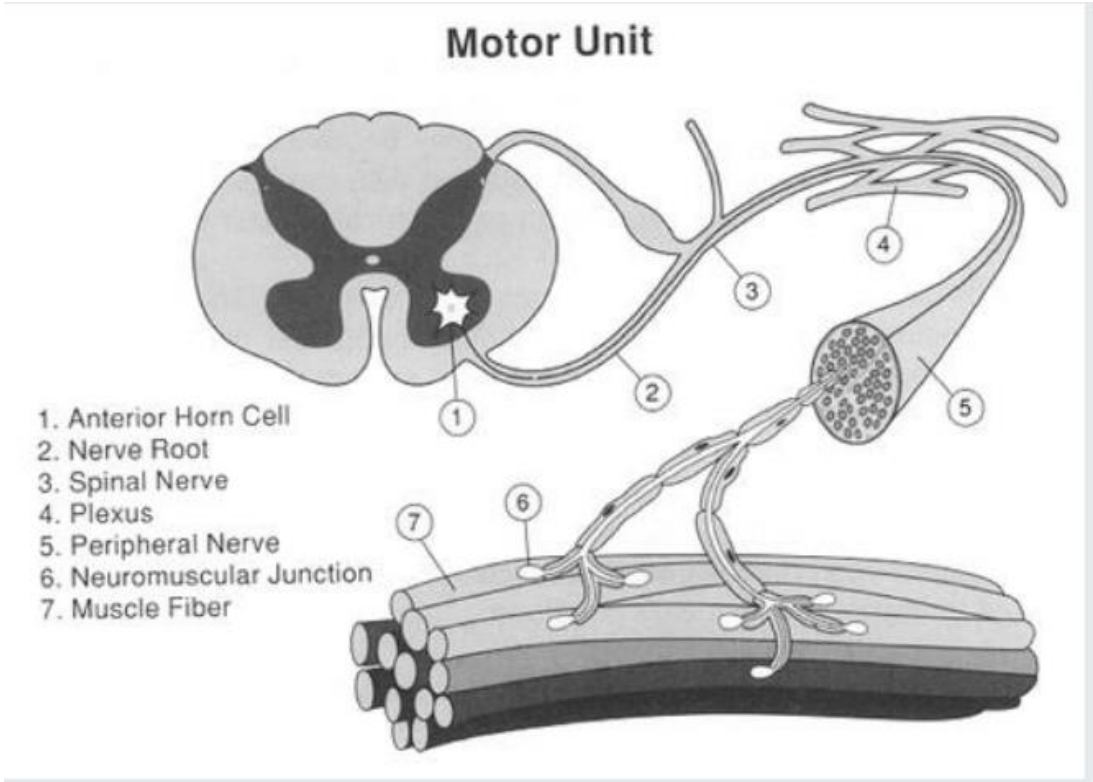
By the end of this session, students should be able to:

- Explain what muscle disorders are, their epidemiology and impact on health.
- Describe the classification of muscular disorders
- Describe the clinical features of common muscular disorders
- Explain the principles of management
- Explain the physical management of muscular disorders



Neuromuscular conditions





1. α-motor neurone,
2. Nerve roots
3. Spinal nerve
4. Plexus
5. Peripheral nerve
6. Neuromuscular junction
7. Muscle fibre

Classification of neuromuscular disorders

α-motor neurone pathology

- **Poliomyelitis** *Spinal Muscular Atrophy, Motor Neurone Disease (MND/ALS)*

Nerve root

Spinal nerve

Plexus Brachial plexus

Peripheral nerve pathology

- Peripheral nerve lesion – Common peroneal nerve
- Neuropathy - *Guillain-Barré syndrome, CIDP, Charcot-Marie Tooth disease, Diabetic neuropathy*

Neuromuscular junction disorders

- *Myasthenia Gravis, Lambert Eaton Myasthenic Syndrome*

Muscle pathology or Myopathy

- *Inflammatory (myositis),*
- *Degenerative (dystrophies).*
- *Contractile (myotonias)*



Muscle disorders (Muscle dystrophy)

- Group of inherited disorders characterized by degeneration of skeletal muscle

- Rare
- Progressive
- Physical disability
- Reduced life expectancy
- Prevalence: 1: 1000
- Same age presentation
- Improvement in management
- Longer survival – older patients with MD

https://www.musculardystrophyuk.org/conditions/what-is-muscular-dystrophy/

Types and classification of Muscle Diseases

- Advances in genetics and molecular biology
- Genetic testing has broadened ability to detect different types
  - BMD
  - LGMD
- Muscular dystrophy (MD) – heterogenous group of genetically determined disorders associated with progressive degeneration of skeletal muscles
- Classified according to age of onset, inheritance pattern, and molecular and genetic defect
- Dystrophin protein defect
- X-chromosome inheritance
- Males are affected but females are carriers

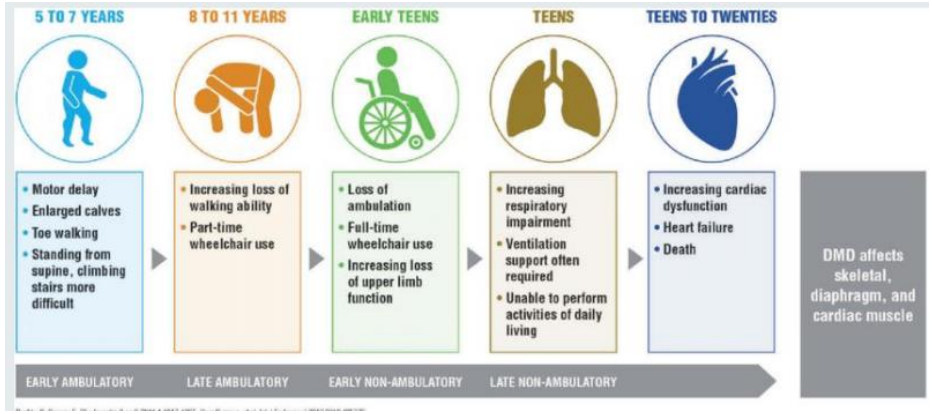
- Duchenne muscular dystrophy (DMD)
- Becker muscular dystrophy (BMD)
- Congenital muscular dystrophy (CMD)
- Emery-Dreifuss muscular dystrophy (EDMD)
- Facioscapulohumeral muscular dystrophy (FSHD)
- Limb girdle muscular dystrophy (LGMD)
- Myotonic dystrophy (DM)
- Oculopharyngeal muscular dystrophy (OPMD)
- Acquired muscle disorders:
  - Polymyositis
  - Dermatomyositis
  - Myastenia gravis

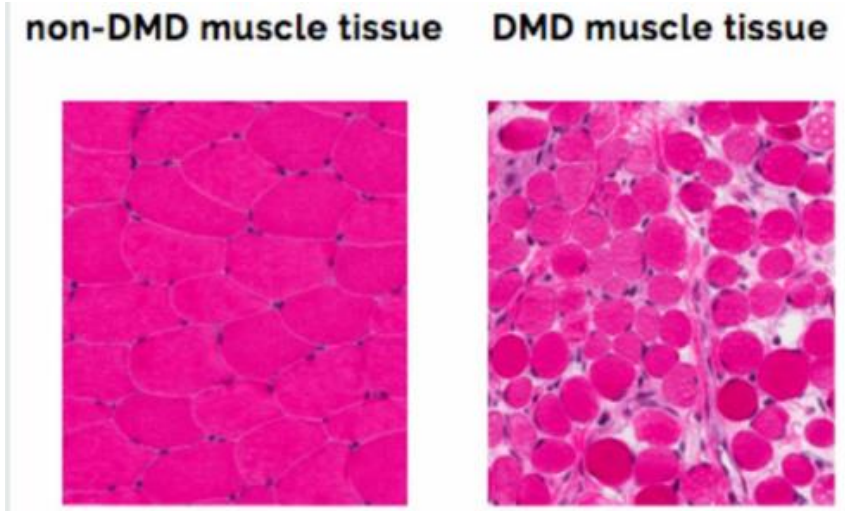
Diagnosis

- History
  - Functional difficulties e.g. frequent tripping, legs giving way, difficulty climbing stairs
- Examination
  - Muscle hypertrophy or atrophy or both
  - Muscle weakness – facial, axial, proximal, distal, or global muscles
  - Muscle contractures
  - Exercise induced myalgia – metabolic or mitochondrial myopathy
  - Respiratory muscle involvement – reduced FVC
  - Systemic complications e.g. early onset cataracts, cardiac rhythm disorder
  - Regular chest infections
- Investigations
  - Serum creatinine kinase (CK) - ↑
  - Biochemical analysis
    - Lactate, carnitine, and acyl carnitine
  - Functional exercise testing
  - EMG
  - Nerve Conduction Velocity
  - Muscle biopsy – definitive diagnostic test for most MD
  - Blood test for DNA analysis may also be done

Duchenne Muscle Dystrophy (DMD)

- First described by Meryon in 1852, later by Duchenne
- Rapidly progressive
- Most severe form of MDs
- 1: 3500 male births
- Prevalence of 1.9 -4.8 per 100,000





Clinical and diagnostic features

- Delayed motor milestones
- Leaning difficulties (1/3)
- Late walker – 18 months
- Diagnosis often missed until weakness starts
- Hip and knee extensor weakness – Gower ‘s maneuver
- Weak abductors – waddling gait
- Wide base of support
- Hip abduction contractures (Iliotibial tract tightness)



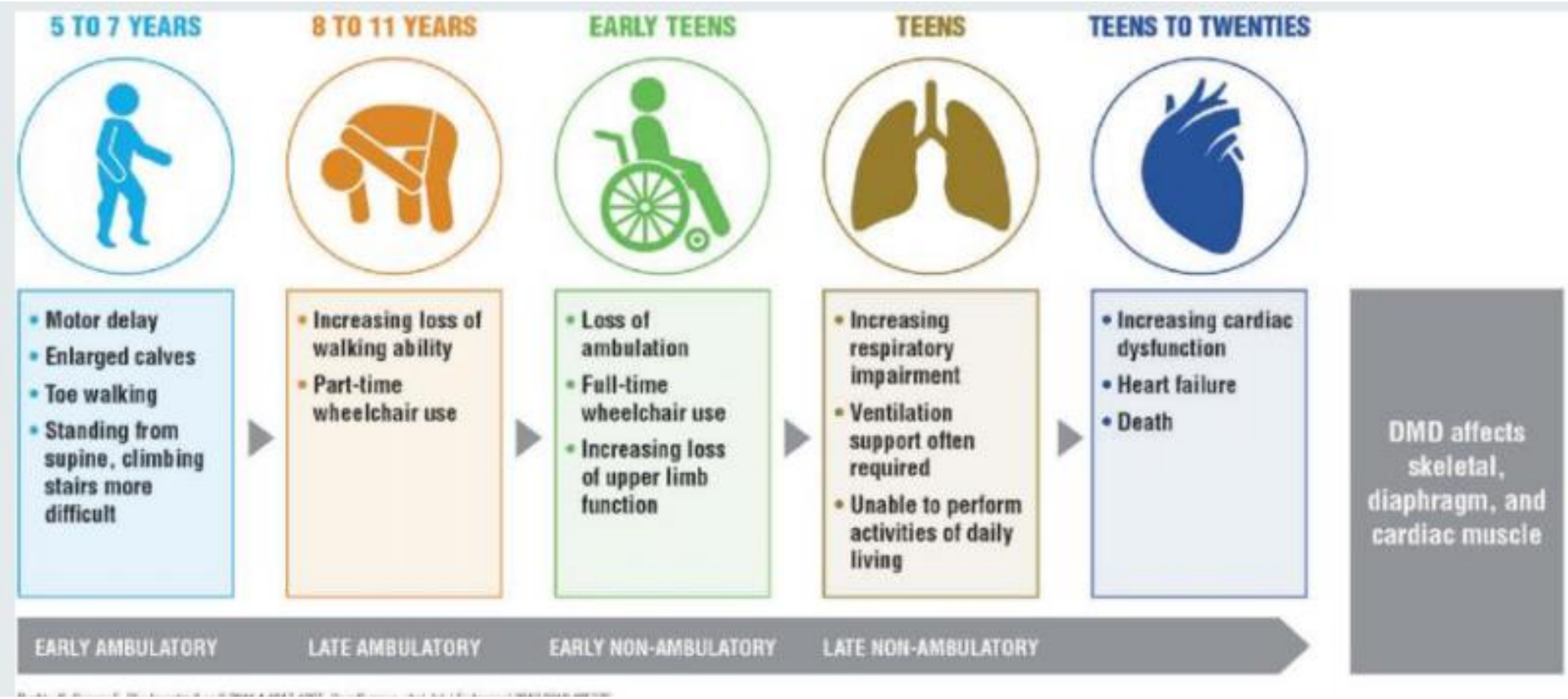
DMD – clinical and diagnostic features

- Toes walking (7-8 years old)
- Equinus foot posture →
- Contracture of Achilles tendon
- Inability to ambulate by 13th birthday
- Flexion contractures elbow, hips and knees
- Bilateral muscle involvement
- Proximal >> distal
- Tendon reflex→ depressed → lost

- In ambulatory phase
  - Pelvic >> shoulder girdle
  - Extensors >> flexors

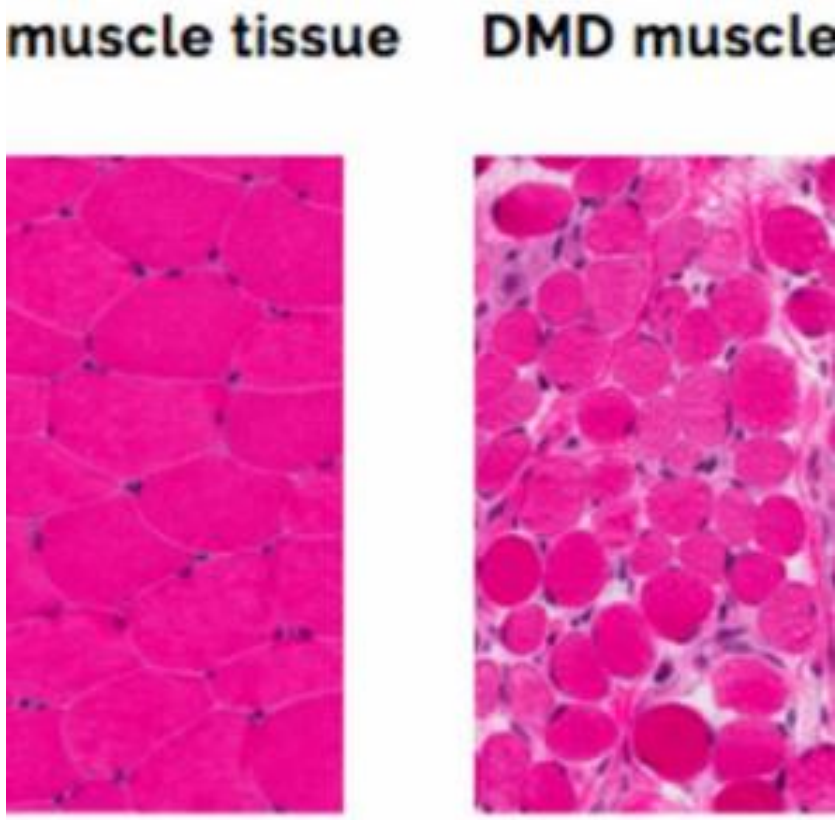
- Contractures
  - Muscle imbalance and postural maladjustment
  - Fixed
  - Scoliosis → ↑ respiratory muscle weakness → bed bound

Progression of DMD



Pathology of DMD

- Weakness → gradual loss of functional muscle fibres with fat and connective tissue due to lack of dystrophin
- Dystrophin - a protein encoded by Xp21 gene is the primary biochemical defect
- Dystrophin - is integral within a complex of proteins which stabilises the integrity of the sarcolemma membrane
- Absence results in a reduction in permeability of the muscle cell membrane
- This leads to excessive quantities of calcium to accumulate within the muscle causing breakdown of myofibrils breakdown and muscle fibre degeneration



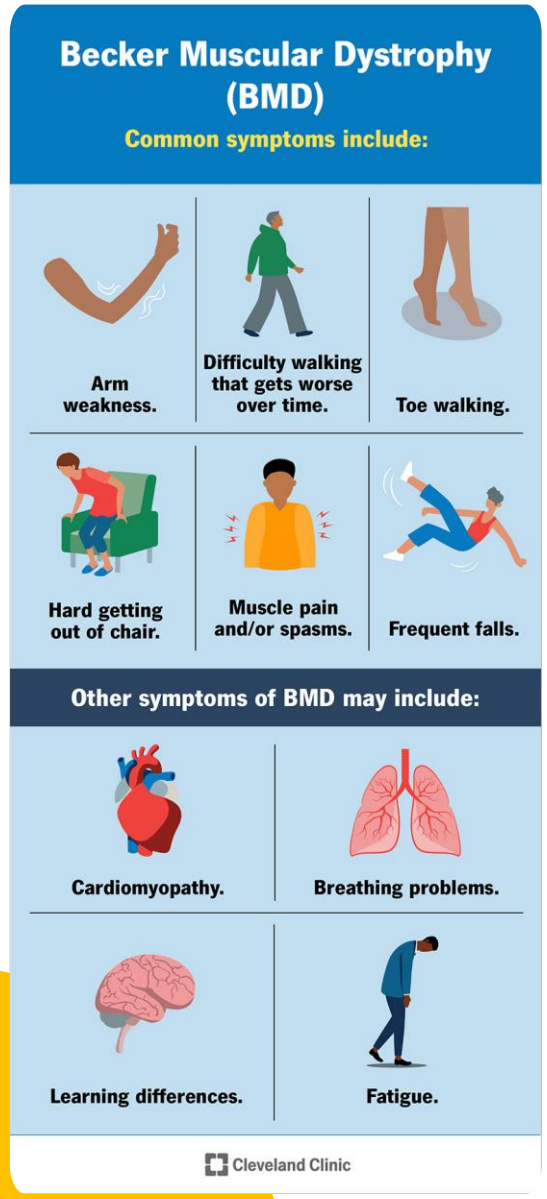
https://youtu.be/DGOmN6rnsNk

Diagnosis

- ↑serum creatinine kinase

- Muscle biopsy – dystrophic features
  - ↑ fibre size variation
  - Evidence of necrosis
  - ↑ central nuclei
  - Hyper contracted eosinophyllic hyaline fibres
  - ↑ fat and connective tissues

- DNA testing (serum blood)
  - Deletion within the dystrophin gene
  - Confirmation of the diagnosis in carriers



Becker Muscular Dystrophy (BMD)

- First described in 1956
- Allelic to DMD but milder
- Only affects males
- Prevalence of 1 in 30000
- Caused by partial deficiency of protein dystrophin
- Symptoms
  - Muscle weakness
  - Muscle atrophy
  - Disability
  - Cardiomyopathy
  - Respiratory muscle weakness

Clinical & diagnostic features of BMD

- Wide spectrum
  - Severe: lost ambulation by 16 years of age
  - Mild: non-progressive cramps and myoglobulinuria
  - Muscle weakness – distribution like DMD
  - Slower progression
  - Less severe contractures
  - Calf pseudo-hypertrophy
  - Ambulation lost – 40%
  - Malignant hyperthermia from general anaethesia
  - Cardiomyopathy – ECG & echocardiogram abnormalities - 50%

- Diagnosis is confirmed by:
  - Genetic mutation in the dystrophin gene
  - Abnormal dystrophin staining on a muscle biopsy

- General Management
  - Prevention of contractures
  - Prolonging ambulation
  - Low intensity aerobic exercises
  - Home adaptation
  - Weight bearing exercises for wheelchair bound patients
  - Prevention of respiratory infections
  - Monitoring od respiratory function - overnight oximetry
  - Non-invasive ventilation
  - Regular cardiac monitoring – yearly ECGs and cardiac echocardiogram

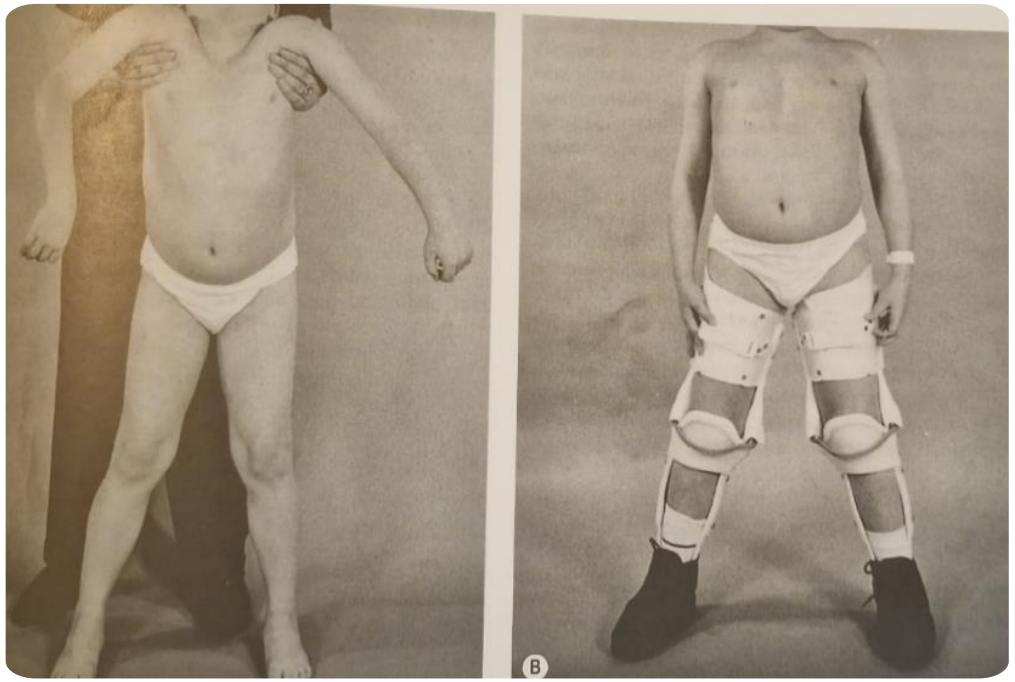
Principles of Management

- Identifying patients' specific needs
- Monitoring complications of the disorders such as cardiac & respiratory monitoring
- Drug therapy e.g. Corticosteroids (muscle strength) but with side effects noted.
- Family management important through an integrated MDT
- Community therapists to provide support and home & school adaptations

General management

- Longer term management
  - Specialist MDT – community medical, educational, social worker
- At diagnosis
  - Support worker to maintain contact
- MDT developmental assessment
  - Screen for learning difficulty
  - Social services to assess state financial support – home adaptation etc
- Physical management
  - Tightness - daily passive stretches of TA with night splints; iliotibial band tightness, hip flexors
  - Prolong ambulation - to prevent rapid development of scoliosis and joint contractures

General management



- At the point of loss of ambulation
  - Light weight Knee Ankle Foot Orthosis (KAFO)
  - Surgery to release tight TA
  - Serial casting
  - Corticosteroids
  - Standing Frame
  - Rigid AFO
  - Wheelchair use
  - Swimming and hydrotherapy
  - Surgery to correct scoliosis

Management of restricted participation

- Electric wheelchair
- Home adaptations
- Ground floor bedroom with shower, bathroom and adapted toilet
- Electric bed
- Adapted motor vehicle
- POSSUM system (Patient Operator Selector Mechanisms)
  - https://www.possum.co.uk/products/qwayo-2/
- Access to internet
- Electronic games



Cardiorespiratory problems

- Restrictive deficit
- Reduction in total lung capacity
- Chest wall stiffness, recurrent aspiration and ineffective cough → recurrent infection
- FVC is a reliable measure of respiratory function
- When it reaches 1 L, average life expectancy without treatment is 3 years
- Treatment by non invasive nasal ventilation

- There is evidence of cardiomyopathy y 18 years
- Cardiac arythmias
- Congestive heart failure
- Monitoring with echocardiogram is recommended once ambulation stops
- Early treatment with ACE inhibitors shown to be beneficial

Final llness

Close monitoring

Liaison with GP, palliative care services

Children Respite Hospices could support family and the boy during this time

Patient support groups e.g. Muscle Dystrophy Campaign

Physical Management

- Approach: Problem solving is preferred
- Treatment planning should involve multi-disciplinary team (MDT),patient, parents and carers

- Assessment
  - Measurement of muscle strength
    - Manual muscle testing
    - Dynamometry
  - Measurement of joint range
    - Goniometry
  - Measurement of functional performance
    - Ability to rise from the floor
    - Transfers
    - Sit to stand
    - Bed mobility
    - Gait Analysis
  - Timed tests
    - 6-minute walk test
    - 10 metre walk test
    - Timing the Gowers' manoeuvre
  - Lung function test
    - Spirometry

Treatment principles

Maintain	Maintain muscle strength and retard contracture progression
Prolong	Prolong ambulation with orthosis
Delay or control	Delay or control scoliosis
Treat	Treat any respiratory and cardiac complications

Social and psychological issues in neuromuscular disorders

Pre-school years

- Support for the family during diagnosis
- Education about the future trajectory of the MD

School Years

- Optimum communication between school, medical team, parents,carers
- Facilities at school must afford integration with able-bodied colleagues
- Realistic goals setting taking into consideration all aspect of life

References

- Thompson N & Quinlivan R (2011). Muscle Disorders. Chapter in Stoke M & Stack E Physical Management for Neurological Conditions. 3rd edition. London. Elsevier Churchill Livingstone.